

Retrofitting a commercial RF induction generator into a computer-controlled, vacuum-integrated annealing system for reactive-metal grain growth

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High-temperature vacuum annealing near a metal’s melting point drives controlled grain growth but normally requires an expensive turn-key vacuum induction furnace. We describe an open, generator-agnostic modernization stack that converts a bare commercial RF induction generator into a computer-controlled, vacuum-integrated, pyrometer-feedback annealing system: LabVIEW/DAQ analog power control, dual-wavelength optical temperature feedback, a high-vacuum quartz-tube chamber, and an in-house machined graphite crucible/susceptor for metal specimens. The layer transfers between generators that expose a monotonic analog power-control input; the reproducible build is anchored on a current 6 kW solid-state generator. In operation the system shows a fixed-geometry power–temperature calibration ($R^2 = 0.991$ over 1200–1400 °C), repeatable 1200 °C / 12 h nickel anneals (1201.2 ± 1.3 °C across 8 runs), stable soaks up to 40 h at 1325 °C, and annealed-nickel microstructures characterized by optical metallography, SEM, and EBSD. A tantalum-susceptor variant extends the furnace to non-coupling ceramics at 2000–2350 °C, demonstrated with YSZ grain growth. Complete design files, bill of materials, control software, and data are openly available.

Keywords: induction furnace; grain growth; vacuum annealing; LabVIEW; pyrometer; open-source hardware

I. INTRODUCTION

High-temperature annealing near a metal’s melting point drives controlled grain growth, central to studying microstructure–property relationships in metals such as nickel and iron^{1–5}. Doing this without oxidizing the sample requires both high temperature (~ 1400 – 1500 °C) and high vacuum or an inert atmosphere^{6,7}. Commercial vacuum induction furnaces meeting these requirements are closed, turn-key systems costing \$50k–\$200k+; resistive tube and box furnaces reach lower temperatures with slower ramps and offer no susceptor route to non-coupling charges; and bare industrial induction generators are widely available but ship without vacuum, optical temperature feedback, or computer setpoint control.

This work adds exactly that missing layer: a bare RF induction generator^{8,9} is retrofitted into a full annealing system with computer power control, closed-loop optical temperature feedback, a high-vacuum quartz-tube chamber, and a machined graphite crucible/susceptor. The stack is deliberately generator-agnostic: the only requirement is an analog power-control input (e.g. 4–20 mA or 0–5 V) with a monotonic command→power response, since the pyrometer feedback and PID calibration absorb any nonlinearity. The generator internals are not reproduced; a reader supplies a compatible generator and rebuilds the layer described here. The graphite crucible, which couples strongly to the RF field, serves the metal grain-growth charges; for non-coupling ceramics such as YSZ, an entirely separate tantalum-susceptor charge stack replaces it (Sec. V A).

The reproducible build documented here is anchored on a current, USA-sourced 6 kW solid-state generator whose controller accepts a standard 4–20 mA power

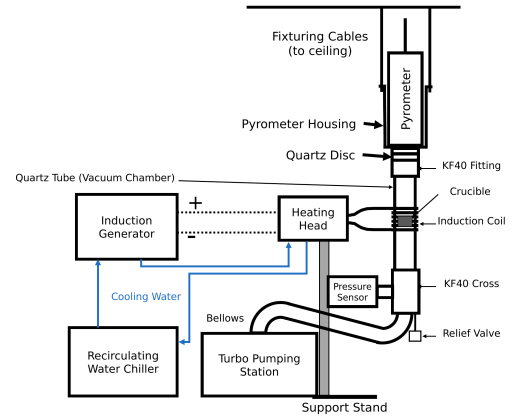


FIG. 1. System overview. The generator drives the water-cooled heating head/work coil around the vertically suspended quartz-tube chamber; the chamber stack is joined by KF40 flanges to the ceiling-suspended pyrometer housing, and an overpressure relief valve and flexible bellows lead past the pressure sensor to the turbo pumping station.

setpoint. The generator + heating head + controller + chiller package was quoted at $\approx \$16.5$ k, with the retrofit components itemized in the supplementary bill of materials — a small fraction of turn-key cost, in a form that is repairable and modifiable by the user. All design files, control software, run logs, and characterization data are openly available (see Data Availability), with hardware under CERN-OHL-S v2 and code under MIT (proposed).

II. SYSTEM DESIGN AND DESCRIPTION

The system comprises five subsystems (Fig. 1); Fig. 2 shows the assembled system at operating power.

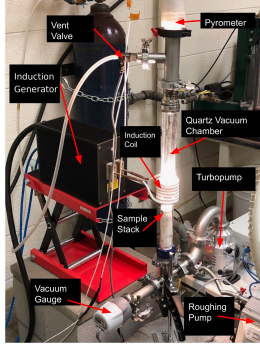


FIG. 2. The assembled, functioning furnace with the main components labeled; the quartz chamber glows at the sample-stack position, the pyrometer sights down the chamber axis, the vent valve admits the argon backfill, and the gauge, turbopump, and separately supported roughing pump complete the vacuum path.

The heating subsystem is a 6 kW solid-state induction generator driving a water-cooled copper work coil (~ 3 in tall, 2.5 in ID, ~ 6.5 turns, tuned to the sample/crucible stack^{10,11}); the coil conductors carry both the RF current and the coolant, and cooling water flows in series from the chiller through the generator and then the heating head/work coil before returning to the chiller. The vacuum and gas subsystem starts from the 35 mm-ID fused-quartz tube that forms the chamber, connected through a KF40 flange and flexible bellows to a compact turbo pumping station ($\sim 1 \times 10^{-6}$ – 10^{-8} Torr); a manual vent valve fed from an inert-gas regulator allows controlled venting and backfilling^{12,13}. The atmosphere matters for graphite-crucible anneals: the best results were achieved by pumping to high vacuum and then backfilling with argon through the vent valve, purging before settling to a steady trickle (~ 20 SCCM), whereas heating under vacuum alone tended to sublime graphite onto the chamber walls; for the highest-temperature ceramic work (Sec. V A), the chamber was evacuated and then continuously filled with argon while the turbopump was running, avoiding oxidation during heating.

Temperature is sensed optically: a dual-wavelength (ratio) pyrometer^{14,15} (800–2500 °C sensing range) views the sample through a KF40 optical window, providing a non-contact signal robust to emissivity changes and to the RF field that would corrupt thermocouples^{16–18}. Control is implemented in LabVIEW, with VIs for manual control, automated ramping, PID tuning, and email alerts: the DAQ’s 0–5 V analog output drives a voltage-to-current loop conditioner whose 4–20 mA output lands on the generator’s power-setpoint input, and the pyrometer’s analog output returns to a DAQ input, closing the PID loop. Below the pyrometer’s

~ 700 °C detection floor the VI ramps open-loop; once the pyrometer returns a sustained valid reading, control hands over to closed-loop PID on the optical temperature^{19,20}. Mechanically, a bolted support stand carries the generator’s heating head, centering the work coil on the crucible position; the vacuum chamber itself has no support stand — the chamber stack is joined by KF40 flanges to the pyrometer housing, suspended from the ceiling by cables, so the vertical stack hangs from above.

Metal charges in the graphite crucible reach the Ni/Fe grain-growth range (~ 1400 – 1500 °C) with demonstrated soaks from minutes to ~ 40 h, and the separate tantalum-susceptor ceramic configuration extends the working range to ~ 2000 – 2350 °C. The layer transfers to any generator that exposes a monotonic analog power input, supports a water-cooled head/coil of the documented geometry, delivers enough RF power for the target charge, and allows electrical isolation of the DAQ/control path; when porting, the command scaling, calibration, PID gains, alignment, and interlock behavior are re-established — already demonstrated by moving the layer between two generators of very different designs.

A. Graphite crucible / susceptor

Metal specimens are held in a two-piece graphite crucible machined in-house from 3000 °C-grade graphite stock: a cup body and a lidded cap with a central bore that contains the charge while leaving an optical path for the pyrometer. The graphite couples strongly to the induction field and acts as a susceptor, heating the contained metal specimen by conduction and radiation. The specimen is sandwiched between two alumina discs in the cup, the lid seats on top, and a sapphire window closes the lid bore (Fig. 3); fully assembled the stack stands 13 mm tall, and the measured dimensions of every piece are given in the supplementary material. Inside the vacuum chamber the crucible rests on an alumina support rod that positions it at coil height, and crucibles are repaired with graphite cement rather than discarded. The graphite crucible is used exclusively for metal grain growth; ceramic charges are never loaded into it, and instead use the entirely separate tantalum-susceptor stack of Sec. V A.

III. CONSTRUCTION AND OPERATION

Following the heating path, the generator’s heating head mounts on an adjustable platform on the bolted support stand so the coil centers on the crucible position of the vertically suspended quartz tube. The vacuum path begins at the quartz tube, whose open end terminates in a KF40 flange closed by a 55 mm quartz optical disc; every flanged joint compresses an elastomer O-ring carried on a centering ring, so the

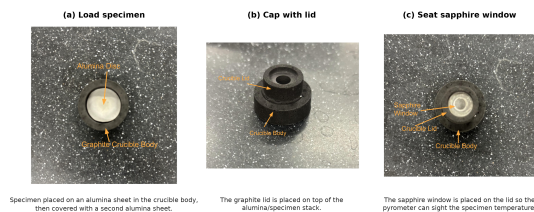


FIG. 3. Specimen-loading sequence for the graphite crucible/susceptor. (a) The specimen is placed between two alumina sheets in the crucible body. (b) The graphite lid is seated on top. (c) The sapphire window closes the lid bore, leaving an optical path for the pyrometer.

stack assemblies with hand-tightened clamps and no tools, and these seals support the $\sim 10^{-6}$ – 10^{-8} Torr operating vacuum. From the bottom of the stack, a tee carries a 0.5 psi overpressure relief valve protecting the quartz tube during backfilling, and a flexible bellows continues through the wide-range vacuum gauge to the turbo pumping station; the roughing pump sits on a separate support, mechanically decoupled so its vibration does not reach the chamber or the pyrometer sight line (photographs in the supplementary material). On the gas path, the argon regulator feeds the manual vent valve; a mass flow controller between them is optional (a regulator and needle valve suffice). On the control path, electrical isolation is maintained between the DAQ/control wiring and the generator interface, and manual control is verified before automated ramp/soak is enabled.

Operationally, before RF power is applied the chamber must be at high vacuum or backfilled, cooling-water flow confirmed and leak-free, the coil/sample/pyrometer sight line aligned, and conductive objects kept 6–12 in from the coil. A run begins with the chamber vented (never while the turbo pump is spinning): the specimen is loaded into the crucible (Fig. 3), the stack inserted, and the chamber pumped down. The LabVIEW profile ramps open-loop below the pyrometer floor, hands over to closed-loop PID, and runs the programmed soak; afterwards the sample cools under vacuum and the chamber is vented only once the turbo pump has stopped. Off-normal conditions — loss of cooling water, vacuum loss, arcing, or runaway temperature — all call for zeroing the command and removing RF power first, then locking the system out.

IV. PERFORMANCE VALIDATION

The furnace has completed more than one hundred logged anneals of Ni200 and Ni4N5 nickel, YSZ, and Pd thermal-evaporation charges at ~ 900 – 1400 °C, with soaks from minutes to ~ 40 h; the supplementary material links each result below to its underlying run log.

For one fixed Ni4N5/quartz/graphite/optical configuration, four steady-state anneals spanning 1200 – 1400 °C give a monotonic relation between soak-

mean analog power command and soak-mean pyrometer temperature, well described by $T \approx 931 + 632 I$ (°C, I in mA), $R^2 = 0.991$ (supplementary material); the offset shifts with charge and optical configuration, so this is an operational calibration for a given setup, not a universal command–temperature law. A Ni4N5 anneal at a nominal 1300 °C / 12 h condition serves as the representative closed-loop run (trace in the supplementary material): the soak held 1302.1 ± 3.0 °C over the 12 h plateau, with a 10–90 % rise time of ~ 1135 s, overshoot of ~ 12 °C, and near-zero drift (-0.02 °C/h).

Eight Ni4N5 anneals at a nominal 1200 °C / 12 h condition, run under the same gas-flow configuration, reproduced their soak temperature to 1201.2 ± 1.3 °C (coefficient of variation 0.11 %). Long-duration stability was demonstrated with two Ni200 anneals at 1325 °C, held for 20 h (1326.5 ± 1.5 °C, drift -0.004 °C/h) and 40 h (figures in the supplementary material). Reported temperatures are closed-loop ratio-pyrometer readings in a fixed optical geometry; absolute specimen temperature still depends on emissivity and surface condition^{16,21,22}.

V. MICROSTRUCTURAL VALIDATION

The furnace exists to change microstructure, so the decisive test is what the anneals do to real specimens. Annealed nickel goes straight from the furnace into the electron microscope: Fig. 4 shows a raw electron-backscatter (Kikuchi) pattern from a furnace-annealed nickel specimen, reproduced exactly as the EBSD detector saved it during orientation mapping — even at the coarse 8×8 binning used for mapping, the bands and zone axes are crisply resolved, and inverse-pole-figure maps built from such patterns resolve hundreds of indexed grains per specimen (supplementary material). Thermal grooving is itself a record of the anneal: edge-on views of a specimen annealed at 1200 °C / 12 h, cleaned only ultrasonically, show deep grooves and boundaries traversing the full sheet thickness, a step toward columnar grain structures²³.

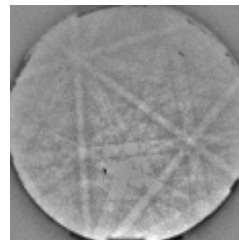


FIG. 4. Raw electron-backscatter (Kikuchi) pattern from a furnace-annealed nickel specimen, exactly as saved by the EBSD detector during orientation mapping (the detector's 8×8 -binned output). The crisply resolved bands and zone axes index directly to the nickel lattice.

Figure 5 shows the microstructure such an anneal

produces, for a Ni4N5 specimen soaked 20 h at 1300 °C (measured soak ~ 1301 °C) under the argon trickle flow: the optical grain map resolves an equiaxed grain network delineated by thermal grooving alone, with grain-size measurements made directly in the image against its 200 μm scale bar, and the SEM detail resolves an individual boundary groove at the micrometer scale²⁴.

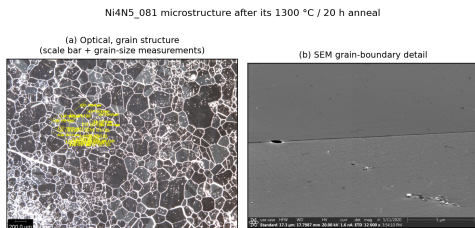


FIG. 5. Microstructure of specimen Ni4N5.081 after a 1300 °C / 20 h anneal under argon trickle flow. (a) Optical grain map with in-image grain-size measurements (200 μm scale bar); (b) high-magnification SEM grain-boundary detail.

A. Extension to ceramic charges: YSZ

The same control, vacuum, and sensing stack extends to charges that do not couple to the RF field at all, demonstrated with grain growth in yttria-stabilized zirconia (YSZ). Ceramic grain growth requires soak temperatures beyond the graphite/alumina stack's service range, so only the charge stack is exchanged: the YSZ specimen is sandwiched between two 25.5 mm tantalum susceptor blocks seated in a ceramic crucible (MgO or boron nitride) on an alumina support rod, while the generator, vacuum path, pyrometer, and control are untouched (schematic and micrographs in the supplementary material). With a single tantalum block the pyrometer reads ~ 1230 °C at a 10 % power command, rising monotonically to ~ 2020 °C at 31 %; two stacked blocks shift the curve upward, hold 2000 ± 3 °C for over an hour at a constant 17 % command, and reach 2344 °C at 33 % before the chamber pressure interlock ended the ramp — the practical ceiling so far is outgassing at temperature, not available RF power, and the pyrometer reads to 2500 °C.

At these temperatures, chemical compatibility between specimen, susceptor, and crucible becomes the design constraint, mapped empirically by operating the furnace across susceptor/crucible combinations: a Ta block atop the YSZ at 2500 °C/8 h grew grains successfully but deposited Ta vapor; YSZ set into a graphite susceptor at 2000 °C/168 h was consumed by the carbon; and a Ta/YSZ/Ta sandwich at 2130 °C/80.4 h formed a Ta–BN reaction shell. Figure 6 shows the outcome of the first fully successful ceramic anneal, 45 min at 2500 °C: the grain-boundary network of the YSZ surface after coarsening from ~ 20 to ~ 90 μm — growth that a commercial 1600 °C box furnace needed more than

228 h to approach (10 to 80 μm). Two practical rules emerged: shorter, hotter anneals stay clean (runs past ~ 3 h contaminated the specimen), and above ~ 2200 °C a carbon susceptor not touching the YSZ is the indicated configuration.



FIG. 6. As-recorded optical micrograph of the YSZ specimen after the 2500 °C / 45 min tantalum-susceptor anneal, which coarsened the grains from ~ 20 to ~ 90 μm . The red multi-grain measurement traces and the 100 μm scale bar are part of the original microscope record.

VI. CONCLUSIONS

A bare commercial RF induction generator can be converted into a computer-controlled, vacuum-integrated, pyrometer-feedback annealing system with a documented, generator-agnostic modernization layer. Running on a current solid-state generator, the stack demonstrated an operational power–temperature calibration ($R^2 = 0.991$), soak repeatability of 1201.2 ± 1.3 °C across eight nominally identical anneals, stable closed-loop soaks up to 40 h, and annealed nickel whose surfaces index directly in the EBSD and show controlled grain growth; a tantalum-susceptor variant extends the same furnace to non-coupling ceramics at up to 2344 °C. Because the layer requires only a monotonic analog power-control input, the design is directly reusable by other laboratories at a small fraction of turn-key cost.

SUPPLEMENTARY MATERIAL

The supplementary material contains the extended figure set, the specimen–run cross-reference (Table S1, hyperlinked to the raw logs and image data), the itemized bill of materials, the design-file inventory, and the standard operating procedure; the repository (see Data Availability) holds the control software, editable schematics, raw and parsed anneal logs, the SEM/EBSD and optical characterization data, and the scripts that reproduce every figure and metric in this paper.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Ethics Approval

This work did not involve human or animal subjects; ethics approval is not applicable.

Author Contributions

— provisional: **Sterling G. Baird:** Conceptualization, Software, Investigation, Writing – original draft. **Ryan Weber:** Investigation, Methodology, Data curation, Visualization. **Christopher Nyborg:** Investigation, Methodology, Data curation, Visualization. **Ronnie Guymon:** Visualization, Writing – review & editing. **Gage Erickson:** Visualization, Writing – review & editing. **Oliver Johnson:** Supervision, Project administration.

DATA AVAILABILITY

The data that support the findings of this study are openly available in the development repository (<https://github.com/vertical-cloud-lab/custom-induction-furnace>); an archival snapshot is deposited at Zenodo, <https://doi.org/10.5281/zenodo.20878017>.

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